

An algorithm portfolio for the dynamic maximal covering location problem

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Abstract The location of facilities (antennas, ambulances, police patrols, etc) has been widely studied in the literature. The maximal covering location problem aims at locating the facilities in such positions that maximizes certain notion of coverage. In the dynamic or multi-period version of the problem, it is assumed that the nodes' demand changes with the time and as a consequence, facilities can be opened or closed among the periods. In this contribution we propose to solve dynamic maximal covering location problem using an algorithm portfolio that includes adaptation, cooperation and learning. The portfolio is composed of an evolutionary strategy and three different simulated annealing methods (that were recently used to solve the problem). Experiments were conducted on 45 test instances (considering up to 2500 nodes and 200 potential facility locations). The results clearly show that the performance of the portfolio is significantly better than its constituent algorithms.

Keywords Coverage · Location · Metaheuristics · Algorithm portfolio

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1 Introduction

Nowadays the location of facilities to satisfy the demand for services is gaining more interest. Over the last decades, a large number of operations research techniques, tools and algorithms have been applied to a wide range of location problems. A detailed introduction of location models can be found in the work by Reville and Eiselt [21]. The interested reader can also check [2,9].

Among the location problems, the so called maximum covering location problem (MCLP) is a classic model [5]. The MCLP problem has many applications in production and services, such as police patrol station locations [4,7,37], location and redistribution of ambulances or emergency units [16], location of public facilities [24], train station location in metropolitan areas [29], or allocation of doctors to primary health care centers [31], among others.

In order to model certain dynamic features of the problems, the authors in [9] proposed the dynamic covering allocation problem (DCAP) where the coverage radius varies over the time. Also, if the facilities are movable (like ambulances), their location can be changed. Otherwise they are located once but the assignment of demand nodes may change through the time. Rajagopalan et al. [19] proposed the Dynamic Available Coverage Location (DACL). The aim of this model is to determine the minimum number of facilities and their locations in each time period in order to satisfy certain covering restrictions.

Also, Gunawardane in [14] proposed a dynamic variant or multiperiod maximal covering location problem (DMCLP), which is claimed to be a more realistic model because it considers various time periods. The main objective in DMCLP is to maximize the coverage over all time periods, allowing for the relocation of facilities.

For solving the DMCLP both exact and approximate methods were proposed. In the former category, integer programming was used to solve a model with only two time periods [24]. Also a linear programming approach was proposed to maximize coverage under different travel times and fleet sizes in two periods [20]. Approximate methods (essentially metaheuristics, like Tabu Search and Simulated Annealing) have been used to solve bigger problem instances (with a greater number of periods and nodes) [12, 36].

Despite the relative success of the previous approaches, it seems that there is a lack of methods able to manage bigger or large scale problem instances. As stated in [2]:

“Since the inherent complexity of facility location problems (specifically dynamic problems) has been troublesome, existing methods are no longer effective in dealing with these problems (particularly large-scale problems). Hence, methods bringing heuristic or metaheuristics solutions have been initiated and established... However, they may not be sufficient in dealing with dynamic problems, and hence, more state-of-the-art hybrid techniques can be applied in upcoming research works.”

One approach in this direction has been recently proposed in [36], but to the best of our knowledge, in said contribution and others, researchers applied isolated or “pure” methods.

The application of hybrid metaheuristics in dynamic facility location problems has not been explored enough, despite their success in solving large scale problems in other domains, like inventory routing [28], supply chain [33], assignment problems [30], graph coloring [23], etc. These hybrid strategies combine the strengths and weaknesses of various individual methods to obtain a combined approach able to obtain a much superior performance than their individual components.

Among hybrid metaheuristics, we focus here on algorithms portfolios, which are defined as a set of different algorithms (or the same algorithm with different settings) that are run in parallel while sharing information about their search process [10, 13, 15].

As a showcase of the success of portfolio approaches for solving complex and large scale problems, we can cite the works by Shukla et al. [27, 28] both in the context of routing problems variants. In the former case, the authors focused on the inventory routing problem with stochastic demands and proposed a portfolio composed of four types of genetic algorithms. In the latter case, the focus was on the vehicle routing problem, also with stochastic demands. Here, the authors proposed a portfolio comprising population based methods (genetic algorithms and artificial immune system) and trajectory or local search methods (namely tabu search and simulated annealing). In both works, the

experimentation showed that the portfolios obtained a higher performance than their individual counterparts when facing complex instances.

On the context of real function optimization, algorithm portfolios have also shown to be competitive. For example, the work presented in [35] proposed a general methodology to automatically compose a “good” portfolio from a set of selected evolutionary algorithms. This methodology was used to construct a portfolio composed of five evolutionary algorithms. The test done over the CEC2005¹ and CEC2013 benchmarks proved that the portfolio obtained better average ranking than five state-of-the-art algorithms.

Another case of success can be found in [34], where the authors applied this type of method to solve the Heating Ventilation and Air Conditioning design problem. Concretely, they proposed a measure to predict which algorithm in the portfolio is the best at any given point in time. From the experimentation done, they concluded that by means of this measure, the portfolio was able to select the most appropriate algorithm for the instance at hand, and that its performance was better than choosing an algorithm randomly.

Algorithm portfolios have been also applied in complex boolean optimization problems like SAT (Satisfiability Problem) [32], Max-SAT (Maximum Satisfiability Problem) [1], QBF (Quantified Boolean Formula) [18], and CSP (Constraint Satisfaction Problem) [17], where this sort of method has become state-of-the-art, ranking in top positions in most of the competitions related to these problems.

So, given the relevance of dynamic versions of location problems and the demonstrated success of portfolio approaches for solving complex optimization problems, this paper is aimed at solving large scale versions of the DMCLP proposed in [14], using a portfolio of algorithms comprising (among others) the isolated methods proposed in [36]. The portfolio also includes adaptation, cooperation and learning. A similar portfolio was previously presented in [8], to solve synthetic dynamic optimization problems with excellent results. Now, in order to evaluate our proposed portfolio’s performance: (1) we used the same test instances and scenarios defined in [36] and (2) we compared it with the three algorithms from [36].

The article is organized as follows. Section 2 reviews the literature related to DMCLP. In Sect. 3 we define the problem and the representation used. Section 4 describes the algorithm portfolio proposed. Then Sect. 5 shows the experimental framework of this work, including the details of the experimentation, the algorithms used in comparisons and non-parametric statistical tests used for the evaluation of results. The next part of the article in Sect. 6 is devoted to the

¹ CEC stands for Conference on Evolutionary Computation, where competitions over well-defined optimization problems’ benchmarks are done.

analysis of the results. Finally, we provide the conclusions in Sect. 7.

2 Related work on covering location problems

In the context of location problems, we can find the so called covering location problem (CLP) [5] where the aim is to cover a set of demands bearing in mind different objectives to be optimized. The two best known categories of CLPs are: maximal covering location problem (MCLP) [5] and set covering location problem (SCLP) [14]. In both categories a demand is covered if it has a facility located at a time/distance equal to or lower than a predefined threshold. SCLP aims at covering all demand nodes with the least possible number of facilities, whereas MCLP aims at covering the maximum possible demand with a known number of facilities. Here we focus on MCLP.

MCLP was first introduced by Church and ReVelle in [5]. The aim of the model is to maximize the coverage of a population with limited resources. The MCLP considers a discrete set of demand nodes and a discrete set of nodes where facilities can be located. Each node has an associated demand, representing their level of importance, and some nodes can remain uncovered. The objective of the MCLP is to maximize the demand coverage by finding the best location for a number p of facilities.

The parameters and variables that define MCLP are:

Parameters

- i, I the index and set of demand nodes.
- j, J the index and set of eligible facility locations.
- a_i population or demand for the node i .
- d_{ij} the shortest distance (or time) from demand node i to the facility j .
- S the minimum distance (or time) between a demand node and facility to be considered as covered.
- $N_i \{j \mid d_{ij} \leq S\}$ set of potential facilities that can cover the demand generated in i .
- p number of facilities to be located.

Variables

- $X_j \{0, 1\}$ a binary variable which equals 1 if the facility is placed at node j , 0 otherwise.
- $Y_i \{0, 1\}$ a binary variable which equals 1 if the demand node i is covered by one or more facilities located within a distance of S , 0 otherwise.

The objective function of the problem is defined as:

$$\text{maximize } f(x) = \sum_{i \in I} a_i Y_i \quad (1)$$

Subject to:

$$Y_i \leq \sum_{j \in N_i} X_j, \quad i \in I \quad (2)$$

$$\sum_{j \in J} X_j = p \quad (3)$$

Constraint 2 states that a demand node is covered only when one or more facilities are located within a distance S from this node. Constraint 3 ensures that the number of facilities to be located is p .

In what follows, we briefly review dynamic approaches of the MCLP.

One of the first variants with dynamism was presented in [24]. The authors propose a multiobjective dynamic location problem (MODLP), where every period has associated a specific objective function. Then, these individual values are combined in a single evaluation using a weighted summation. The experiments were performed considering just two periods.

Also, Repede and Bernardo in [20] formulated a multi-period location problem maximizing the coverage provided for time varying travel times and fleet sizes (TIMEXCLP).

A multiobjective model for dynamic relocation of facilities was developed by Gendreau et. al. in [12]. The two objectives are the maximization of coverage and the minimization of the relocation costs. The resulting model is solved by a parallel Tabu Search algorithm.

Schmid and Doerner in [26], developed a multiperiod extension of the DMCLP, called the double standard model (DSM). In this case, a facility is associated with a vehicle that should be relocated in every period to maximize coverage. Just two vehicles are considered. The resulting model is solved by a Variable Neighborhood Search method.

Başar, Çatay and Ünlüyurt in [3], also considered a multiperiod version of DMCLP oriented to the location of emergency units. An interesting feature of the model is the use of a “double coverage” for the demand nodes which is aimed at giving assistance if one facility (ambulance) is not available. The model also allows for a dynamic relocation/redistribution of the facilities within a period. A Tabu Search algorithm is used to solve the problem.

Tanaka in [29] proposed an extension of the MCLP model that is applied and adapted to the railway network in Tokyo, where each period is controlled independently. More specifically, they consider the maximum flow-covering start time location and service problem (MFCLSTP), that is considered an extension of the original MCLP. Two variants of the problem are presented: MFCLSTP1 (the opening time of every metro station is independent) and MFCLSTP2 (all the stations open at the same time). To solve the problem they use an ad-hoc local search algorithm.

Then Schmid in [25] formulate a dynamic stochastic model for planning and delivering ambulances, also allowing for relocation. In order to get a closer to reality model they consider explicitly time-dependent information and variations (throughout the day) with respect to changing request rates and travel times. They used approximate dynamic programming as a solution method.

Finally, Zarandi et al. in [36] propose a solution for the DMCLP proposed in [5]. They compared results between simulated annealing with the commercial tool CPLEX version 12.1. The experiments were conducted in three instances (with 1800, 2000 and 2500 demand nodes) of the problem considering five periods, different number of facilities and covering radius. The results were compared in terms of time and quality metrics of the best solution found. As the size of the instance grew, the simulated annealing approach obtained better results than CPLEX. Although CPLEX is an exact method, the authors stopped it after a certain time.

As we have seen, the review of the literature shows that dynamic models of the MCLP have received increasing attention in recent years. However, we have also checked that only a few articles have addressed large-scale instances of this problem and none of them have employed hybrid metaheuristics to solve it. Unlike these works, in this paper we are going to present a hybrid metaheuristic, concretely an algorithm portfolio, in order to show that these methods can be a powerful tool in solving large-scale instances of the DMCLP. Considering both aspects, the contribution of this paper is to solve the DMCLP with an algorithm portfolio, attempting to show that the hybridization of metaheuristics can also give very good results in the DMCLP, improving simple metaheuristics.

3 Problem definition

The model to maximize the coverage in multiple periods is known as Dynamic MCLP.

The objective of the multiperiod or dynamic maximal covering location problem (DMCLP) entails finding the optimal way of locating p facilities in T different periods of time. The number of facilities to locate is known a priori but no preference on the number of facilities to be located in each period is available.

The parameters and variables that define the problem DMCLP are: **Parameters**

- i, I the index and set of demand nodes.
- j, J the index and set of eligible facility locations.
- t, T the index and set of time periods.
- a_{it} population or demand for the node i in period t .
- d_{ij} the shortest distance (or time) from demand node i to the facility j .

- S the minimum distance (or time) between a demand node and facility to be considered as covered.
- $N_i \{j \mid d_{ij} \leq S\}$ set of potential facilities that can cover the demand generated in i .
- p number of facilities to be located considering all periods in T .

Variables

- $X_{jt} \{0, 1\}$ a binary variable which equals 1 if the facility is placed at node j in period t , 0 otherwise.
- $Y_{it} \{0, 1\}$ a binary variable which equals 1 if node i in period t is covered by one or more facilities located within a distance of S , 0 otherwise.

The objective function of the problem is defined as:

$$\text{maximize } f(x) = \sum_{t \in T} \sum_{i \in I} a_{it} Y_{it} \quad (4)$$

Subject to:

$$Y_{it} \leq \sum_{j \in N_i} X_{jt}, \quad i \in I, \quad t \in T \quad (5)$$

$$\sum_{t \in T} \sum_{j \in J} X_{jt} = p \quad (6)$$

Constraint 5 forbids the coverage of a demand node j_t , if there is not any facility in a distance less than S to it. Constraint 6 ensures that the number of facilities to be established in all periods T is equal to p . It is clear that a facility may be established in more than a period.

In order to solve this problem using a metaheuristic, it is necessary to provide a way of representing a solution to the problem. One alternative is to use a binary vector representation of size $|T| \cdot |J|$ where each bit represents the status of a facility in the corresponding time period. A value of 1 means that the facility is open and available and 0 otherwise. Figure 1 is an example of a solution representation where the number of periods is $|T| = 3$, the number of facilities $|J| = 3$ and $p = 5$. So, in the first period, facilities are located in node 1 and node 3. In the second period, just one facility is

X_{11}	X_{21}	X_{31}	X_{12}	X_{22}	X_{32}	X_{13}	X_{23}	X_{33}
1	0	1	0	0	1	1	1	0
$t = 1$			$t = 2$			$t = 3$		

Fig. 1 Representation of a solution. $X_{jt} = 1$ indicates that facility i is available in the time period t . In this example there are three potential facility locations, three time periods, and $p = 5$

available at node 3. Finally, facilities are located at nodes 1 and node 2 in the last period.

4 Algorithm portfolio

The original concept of algorithm portfolio has evolved over the years. It is inspired by a simple concept: a collection of different algorithms and/or copies of the same algorithm that are run in one or more CPUs [13, 15]. The main motivation to build an algorithm portfolio is to avoid deciding on a single algorithm to solve the problem at hand. The model uses a learning mechanism which allocates resources to the different methods in accordance to their performance, and then selects the best algorithm at each stage of the search. More formally, the portfolio is composed of a set of metaheuristics $A = \{a_1, \dots, a_n\}$, each a_i being a trajectory or population-based metaheuristic. Every a_i has a single current solution $x_{curr}^{a_i}$ (if we consider a trajectory-based method) or a current population $p_{curr}^{a_i}$ (in population-based metaheuristics).

Algorithm 1 shows the inner working of the portfolio. After the initialization of the algorithms, the method goes into the main loop. Firstly, an a_i is selected and run for one iteration (the definition of what we consider by an iteration is given at the end of this section). The selection strategy employed in this step is the well-known roulette-wheel method, where the portion or selection probability for each a_i is assigned according to its credit w_i . This probability is calculated as:

$$P_{selec}^{a_i} = \frac{w_i}{\sum_{j=1}^n w_j} \quad (7)$$

Algorithm 1: Portfolio Scheme Pseudocode

```

1  $n$  = size of the portfolio;
2 for ( $j = 1$ ;  $j \leq n$ ;  $j++$ ) do
3   | initialize  $a_j$ ;
4 end
5 while (not stop-condition) do
6   |  $a_i \leftarrow$  select an algorithm from  $A$  with probability  $P_{selec}^{a_i}$ ;
7   |  $x_c \leftarrow$  run  $a_i$  for one iteration;
8   | if  $f(x_c) > f(x_{best})$  then
9     |  $x_{best} \leftarrow x_c$ ;
10    | for ( $j = 1$ ;  $j \leq n$ ;  $j++$ ) do
11      | update  $x_{curr}^{a_j}$  or  $p_{curr}^{a_j}$  with  $x_{best}$ ;
12    | end
13  | end
14  | perform credit assignment to  $a_i$ ;
15  | for ( $j = 1$ ;  $j \leq n$ ;  $j++$ ) do
16    | update selection probability  $P_{selec}^{a_j}$ ;
17  | end
18 end

```

There are some steps in the portfolio that are executed differently depending on whether the considered algorithm is trajectory-based or population-based. The most important differences appear in how an iteration is performed, how x_c is obtained and how the current solution or population is updated. Regarding the iteration process and the obtaining of x_c , in trajectory-based algorithms, one iteration corresponds to one application of the neighborhood operator to $x_{curr}^{a_i}$ and the evaluation of the acceptance criterion for the move (e.g., tabu and aspiration criteria in Tabu Search, acceptance probability in Simulated Annealing, etc.). In this case, x_c represents the solution resulting from the application of the neighborhood operator.

In population-based methods, one iteration corresponds to the sequential application of their operators (e.g., selection $\rightarrow$ crossover $\rightarrow$ mutation $\rightarrow$ replacement in Genetic Algorithms). When a crossover operator is applied, only one of the two individuals obtained (randomly selected according to a uniform distribution) is considered for the next step of the sequence of operator applications. In case the method considers replacement, the individual generated replaces the worst parent. x_c corresponds to the individual generated in this process. The update of $x_{curr}^{a_i}$ for trajectory-based algorithms, performed in step 11 of Algorithm 1, entails replacing $x_{curr}^{a_i}$ by x_{best} . Regarding population-based algorithms, in this step, x_{best} replaces the worst individual of the population.

The credit assignment mechanism implemented by the portfolio is, in fact, a learning scheme whose objective is to learn what the best performing method/s for the problem at hand is/are. For this reason, we will also refer to the credit assignment mechanism as learning scheme.

The credit assigned to a metaheuristic a_i at time $g + 1$ is calculated as:

$$w_i(g + 1) = w_i(g) + r_i(g) - l_i(g) \quad (8)$$

where:

- g is the current time.
- $w_i(g + 1)$ is the credit obtained by a_i at time step $g + 1$.
- $w_i(g)$ is the current credit of a_i . The initial credit is set to 0, $w_i(0) = 0$.
- $r_i(g)$ is the reward assigned to a_i . It is a value greater or equal to zero that is determined as a function of the quality of the solution generated by a_i (x_c) and the quality of x_{best} .
- $l_i(g)$ is the penalization assigned to a_i if the generated solution (x_c) is worse than x_{best} . It is defined as follows:

$$l_i(g) = w_i(g) * Q \quad (9)$$

where Q is a penalty term and $Q \in [0, 1]$.

The three main components of the learning scheme are: $w_i(g)$, $r_i(g)$ and $l_i(g)$. The combinations of the three components lead different variants of the portfolio, which were analyzed in [8]. According to that study, the best alternative uses an active penalization scheme (if an algorithm generates worse solutions than the best solution found) and some credit is rewarded if the new solution is strictly better than the best one available. The penalty term is set to 0.9 ($Q = 0.9$) if a_i generates worse solutions than x_{best} , and 0 ($Q = 0$) otherwise. The value $Q = 0.9$ was chosen after empirical observations. In this way, we diminish the credit of a method if it does not produce improvement of the best found solution. Also, $r_i(t) = 1$ if the fitness of the generated solution ($f(x_c)$) is strictly better than $f(x_{best})$.

5 Experimental framework

We describe here the test instances, reference algorithms, comparison techniques and the details of the experimentation we followed to evaluate our proposal.

5.1 Description of the instances

We considered the following features in the generation of DMCLP instances: location of demand nodes, potential locations of facility nodes, demand associated with each node and time periods. We followed the approach recently used by Zarandi et al. in [36] that in turn is based on [22].

First, the locations of nodes I and facility locations J were randomly generated using a uniform distribution in $[0, 30]$ for both x and y coordinates. The distances between the nodes were calculated using the Euclidean distance. Second, the demands of the nodes for each time period (we consider five periods) were randomly assigned using a uniform distribution between $[0, 100]$. Finally, we generated three scenarios with the following combinations of nodes and facilities locations: (1800, 100), (2000, 150), (2500, 200).

For every scenario we considered different numbers of facilities ($p = \{65, 70, 75, 80, 85\}$) and coverage radius ($S = \{4, 4.5, 5\}$) leading to 3 scenarios $\times$ 3 coverage radius $\times$ 5 numbers of facilities = 45 test instances.

We considered 5 different algorithms: the portfolio (PF), simulated annealing exponential (SA-E), simulated annealing hyperbolic (SA-H), simulated annealing linear (SA-L) and evolutionary strategy (ES).

We performed 30 independent runs for every algorithm on every test instance. The stopping criterion for all algorithms was set to 10,000 evaluations of the objective function, that is, each algorithm has available 10,000 evaluations per run. Regarding the portfolio, this means that each of its four component algorithms will have available, on average, 2500

X_{11}	X_{21}	X_{31}	X_{12}	X_{22}	X_{32}	X_{13}	X_{23}	X_{33}
1	0	1	0	0	1	1	1	0

X_{11}	X_{21}	X_{31}	X_{12}	X_{22}	X_{32}	X_{13}	X_{23}	X_{33}
1	0	1	0	0	1	0	1	1

$t = 1$
 $t = 2$
 $t = 3$

Fig. 2 Application of NSS2 Operator. Firstly, a period is selected ($t = 3$). Secondly, one facility is closed X_{13} and one is opened X_{33}

evaluations. Of course, this value will depend on the component's performance.

5.2 Composition of the algorithm portfolio

The composition of the algorithm portfolio is one of the most relevant aspects of its design. In this work, the algorithm portfolio is composed of four methods. Concretely, by three Simulated Annealing methods proposed in [36], because of their good performance in this problem, and by an Evolutionary Algorithm to provide a higher diversity to the portfolio. Table 1 displays the parameter settings of the methods in the algorithm portfolio.

The method NSS2 (neighbourhood search structure) [36] has been used as neighbourhood operator in the Simulated Annealing algorithms, and as the mutation operator for the Evolutionary Algorithm. The method works as follows:

1. Select randomly a time period $t \in T$.
2. Select randomly an open facility k , i.e. $X_{kt} = 1$.
3. Select randomly a closed facility z , i.e. $X_{zt} = 0$.
4. Make $X_{zt} = 1$, $X_{kt} = 0$.

Figure 2 shows an example of the NSS2 usage. The period $t = 3$ is selected. The open facility is $k = 1$ and the closed one is $z = 3$. In the second step, facility X_{13} becomes closed and X_{33} is opened.

In order to construct the initial solution for the methods in the portfolio, a constructive heuristic proposed in [36] was used. In simple terms, each facility becomes open or closed according to a probability that depends on the coverage it can provide. The solution obtained is feasible.

5.3 Statistical assessment of results

Nowadays, it is widely assumed that any comparison among a set of algorithms over a set of problems should be supported by statistical testing. In this article we follow the guidelines proposed in [11] where non-parametric statistical testing is

Table 1 Methods that integrate the algorithm portfolio and their parameter settings

Method	Parameter	Setting
Simulated annealing exponential (SA-E)	Initial temperature t_0 [6]	$t_0 = \frac{\Omega}{\ln \chi_0}$ Ω average deterioration of the objective function χ_0 probability of acceptance
	Final temperature t_f	0
	Exponential cooling scheme	$t_n = \frac{A}{n+1} + B$; $A = \frac{(t_0 - t_f)(N+1)}{N}$; $B = t_0 - A$
	Neighborhood operator	NSS2[36]
Simulated annealing hyperbolic (SA-H)	Initial temperature t_0 [6]	$t_0 = \frac{\Omega}{\ln \chi_0}$ Ω average deterioration of the objective function χ_0 probability of acceptance
	Final temperature t_f	0
	Hyperbolic cooling scheme	$t_n = \frac{1}{2}(t_0 - t_f)(1 - \tanh(\frac{10n}{N} - 5)) + t_f$
	Neighborhood operator	NSS2[36]
Simulated annealing linear (SA-L)	Initial temperature t_0 [6]	$t_0 = \frac{\Omega}{\ln \chi_0}$ Ω average deterioration of the objective function χ_0 probability of acceptance
	Final temperature t_f	0
	Linear cooling scheme	$t_n = t_0 - n \frac{t_0 - t_f}{N}$
	Neighborhood operator	NSS2 [36]
Evolutionary strategy (ES)	Population size	50
	Mutation probability	0.9
	Selection operator	Truncation (20)
	Mutation operator	NSS2 [36]

suggested in situations like the one faced in this contribution (several problems, algorithms and configurations).

Firstly, we will apply Friedman's test to check if significant differences exists among a set of algorithms. Besides this, Friedman's average rank allows for sorting the algorithms in terms of performance. Then in order to evaluate in each case if the best algorithm has a significantly different performance with respect to others, we use Holm's and Finner's post hoc tests for one-to-many comparisons between the best algorithm (as indicated by Friedman's rank output) and the rest, with a significance level of $\alpha = 0.05$. KEEL tool [11] was used for the Friedman test and post-hoc Holm's and Finner's tests.

6 Results

The aims of the experiments were:

- Analyse the performance of the algorithm portfolio versus the isolated version of the four methods that compose it. Here, it is important to highlight that the three Simulated Annealing algorithms can be considered as

reference methods in literature because of their good performance in this problem [36].

- Analyze the internal working of the portfolio, evaluating the evolution of the selection probability of the individual algorithms.

The next part of this section is devoted to commenting on the results obtained.

6.1 Comparison versus isolated algorithms

Here we evaluate the performance of the algorithm portfolio with respect to the individual methods that comprise it. We want to assess if the portfolio is able to take advantage of the strengths and weaknesses of its component methods at every stage of the search leading, in the end, to better results than its individual components. Remember that the three variants of the Simulated Annealing can be considered as reference methods for this problem.

In order to analyze the results, we first compared all the algorithms over all test instances and on each instance individually (using the Friedman test). In all cases the test returned a $p - value = 0.0$, indicating that there are significant differences among algorithms, thus rejecting the null

Table 2 Average ranking provided by Friedman's non-parametric test

Algorithm	Global	Inst-1800	Inst-2000	Inst-2500
PF	1	1	1	1
SA-E	2	2	2	2
SA-H	4	4	4	4
SA-L	5	5	5	5
ES	3	3	3	3
p-value	0.00	0.00	0.00	0.00

The portfolio (PF) is compared against the other algorithms. PF is always the best

Table 3 Holm's post-hoc tests results ($\alpha = 0.05$)

Algorithm	Global	Inst-1800	Inst-2000	Inst-2500
SA-E	>	–	–	–
SA-H	>	>	>	>
SA-L	>	>	>	>
ES	>	>	>	>

PF as the control method. A '>' appears if PF improves significantly to the corresponding algorithm; A '–' indicates no significant differences between both methods existed. (Finner's test provides the same results)

hypothesis. Friedman's test also produces an average ranking for each method compared. These results are shown in Table 2, where the algorithm with best average ranking is marked in bold. In all cases, the best algorithm is the portfolio (PF). It is interesting to note that all the algorithms shows the same ranking independently of the size of the instances. The best individual method over all the problems was SA-E, while the worst one was SA-L. Please note that the only difference between SA-E and SA-L is the cooling scheme. Given a new problem, there is no way to decide, a priori, which cooling scheme should be used. Hard decisions like this are the ones that further motivate the use of algorithm portfolios.

We also applied Holm's and Finner's post-hoc tests, at a significance level of $\alpha = 0.05$, in order to check if the difference in performance between the best method (called the control algorithm or method) against the remaining algorithms is significant or not. These results are shown in Table 3. The control method is the portfolio in all cases. The symbol '>' means that the control method improves significantly to the corresponding algorithm; and '–' indicates that no significant differences exist between both methods. In all cases, Holm's and Finner's provide the same results.

When considering the 45 instances together, significant differences between the portfolio (the method with the best average ranking according to the Friedman's test) and other algorithms were detected.

When the instances are categorized by size, no significant difference exists between the portfolio and SA-E. A possible

explanation is the following: the portfolio is slightly better than the individual methods in each group of instances, but not enough to establish that difference as statistically significant. However, globally, when we group all these improvements, the differences become significant.

Overall, it is clear that the proposed portfolio composed of three versions of SA and one implementation of ES is able to obtain better results than the individual algorithms that comprise it.

Figure 3 shows the percentage of demand covered (over the total demand) for each algorithm, coverage radius (S) and number of facilities to be located (p). Results are shown for the three scenarios considered: 1800 demand nodes and 100 facility locations, 2000 demand nodes and 150 facility locations, 2500 demand nodes and 200 facility locations.

As expected, as the coverage radius increases, the demand covered also increases. A similar situation occurs with the number of facilities available.

In every plot, and considering a specific coverage radius and all the possible values for p , we can observe that the highest coverage percentage corresponds to the portfolio (PF). In most scenarios, it is also confirmed that the second best is SA-E, while the worst alternative is SA-L.

We can observe that, even when considering the results per instance, the portfolio is consistently the best algorithm.

6.2 Algorithms usage in the portfolio

In this section we focus on the usage frequency of the algorithms that comprise the portfolio: simulated annealing exponential (SA-E), simulated annealing hyperbolic (SA-H), simulated annealing linear (SA-L) and evolutionary strategy (ES).

We make the analysis considering intermediate values for coverage radius and facilities to locate, that is, 4.5 and 75, respectively.

Figure 4 shows the accumulated number of selections of each algorithm over all the iterations. Each plot corresponds to each of the instances with 1800, 2000 and 2500 demand nodes. The values correspond to averages over 30 runs and we measure the number of selections at 10 time steps equally distributed.

In the scenario with 1800 nodes and 100 potential facility locations, we can observe that the most frequently selected algorithm during most of the search was SA-E. Also, it is interesting to note that, at the end of the runs, both ES and SA-L were selected the same number of times. However their usage is different during the search, SA-L being more frequently used during the first half of the search.

In the middle size instance, the behavior is different. Up to 5000 evaluations, the three simulated annealing algorithms are equally used. Then, the portfolio makes a lower use of SA-L. As in the previous case, ES is the method least employed.

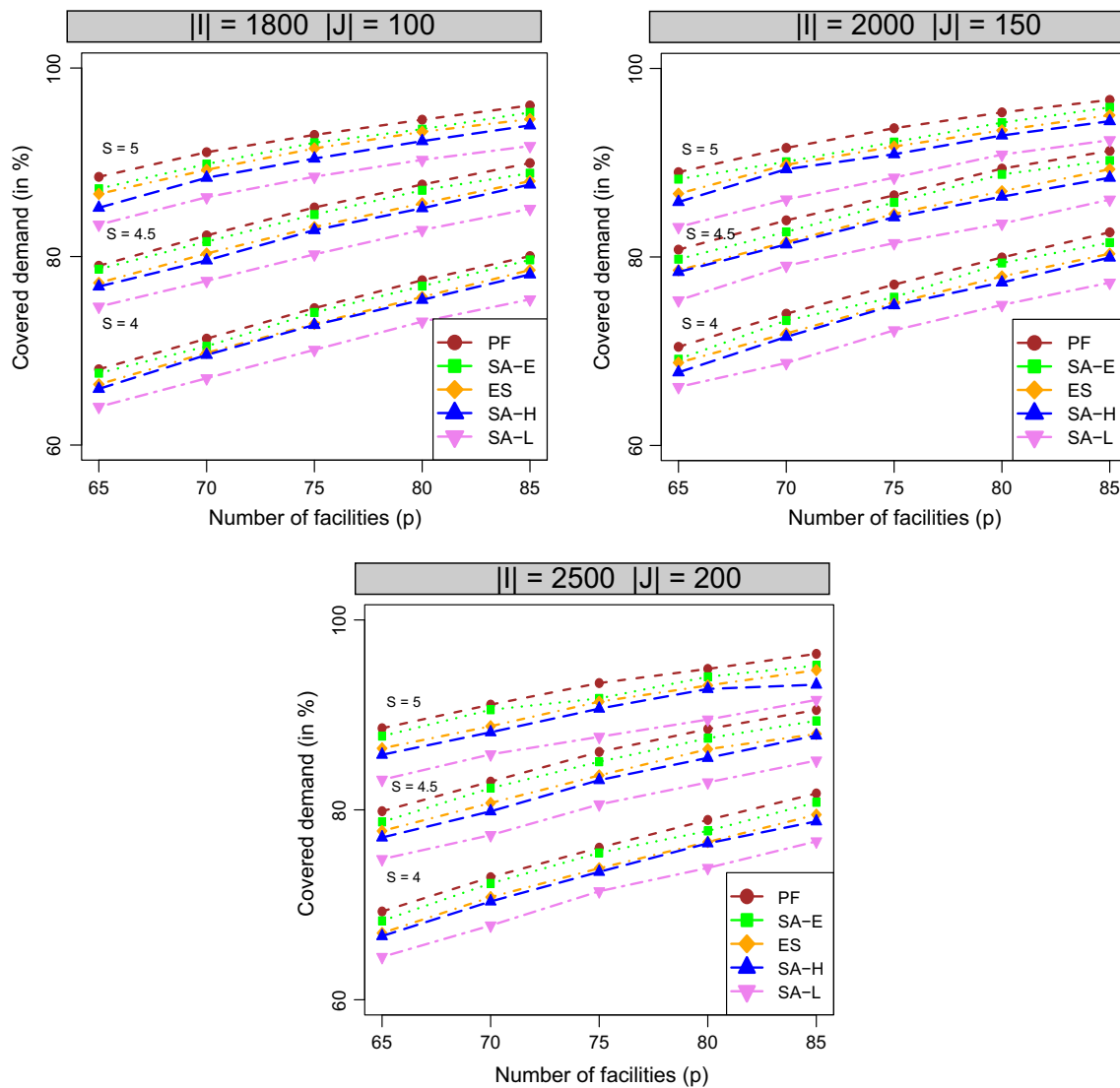


Fig. 3 Percentage of covered demand with respect to the total demand for each algorithm (*Y axis*), coverage radius (*S*) and number of facilities to be located (*p*, *X axis*). Results are shown for the three scenarios con-

sidered: 1800 demand nodes and 100 facilities locations, 2000 demand nodes and 150 facilities locations, 2500 demand nodes and 200 facilities locations

In the biggest instance, we observe that at the beginning, as in the previous case, the three simulated annealing algorithms are equally used. Then, SA-E is clearly more used than the other methods. SA-H and SA-L are employed with the same frequency during all the run.

Finally, it is interesting to compare the behavior of ES in the context of the portfolio and when it is used in isolation. In the former case, it becomes the less selected algorithm while in the latter, it is the second best method. The justification for such behavior is clear and depends on the time that ES has available to be run in each case. When it is used in isolation, it has up to 10,000 fitness functions evaluations available. As it is a population based technique, it has enough time to converge. However, when ES is used in the portfolio, it has

(on average) 2500 evaluations available which is not enough to converge.

7 Conclusions

In this work we have presented an algorithm portfolio to solve the dynamic maximal covering location problem. The portfolio is composed of three Simulated Annealing algorithms (that had shown a good performance in this problem), plus a population based technique.

We did experiments comparing the performance of the portfolio against its individual components. We employed

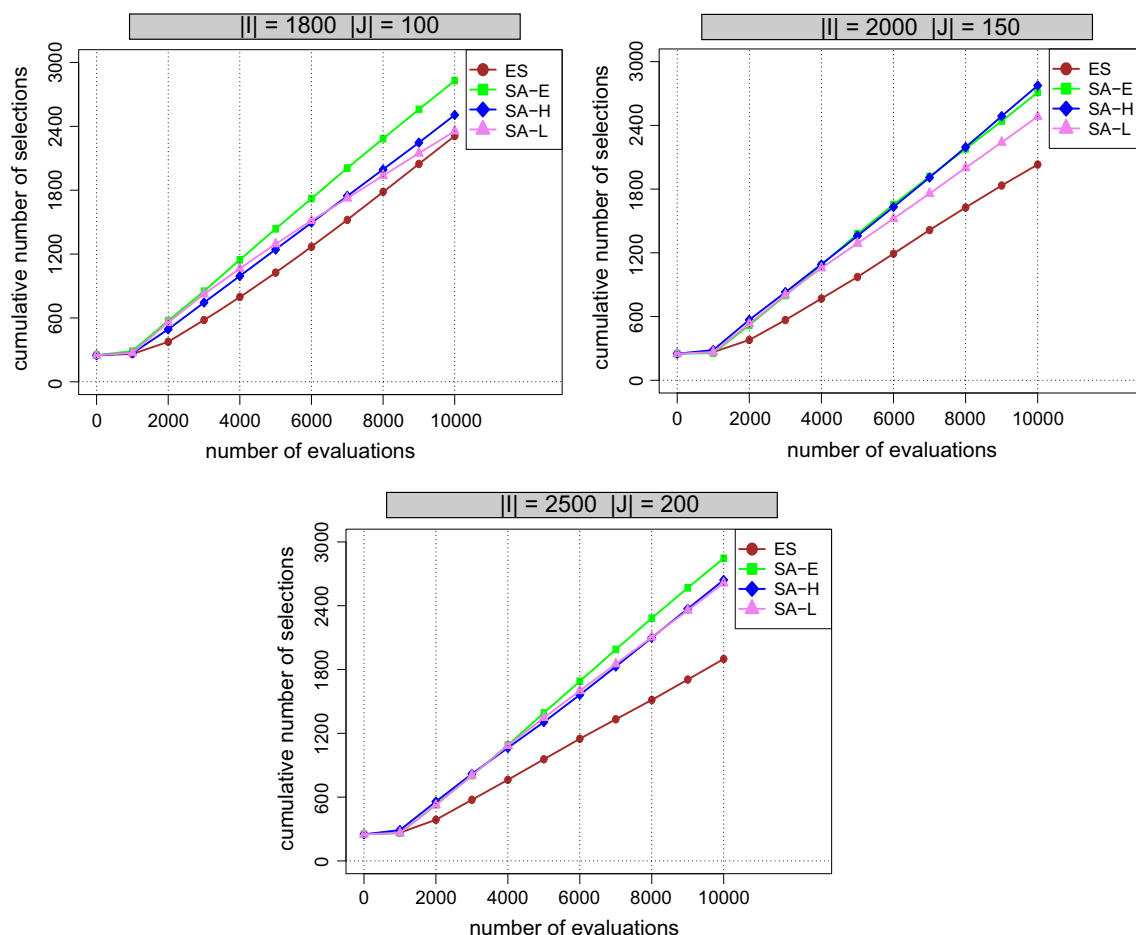


Fig. 4 Accumulated number of selections of each algorithm over all the iterations. Each *plot* corresponds to each of the instances 1800, 2000 and 2500 demand nodes, with respect number of evaluation (*X axis*) and cumulative number of selection (*Y axis*)

non-parametric tests to assess the statistical significance of the differences in performance among the methods.

After analyzing the results of the experimentation we can conclude:

- The algorithm portfolio improved all the individual algorithms significantly when the results over all instances were considered.
- The analysis of the individual algorithms usage revealed that algorithms that do not have a good individual performance (SA-H and SA-L) can be useful at certain stages of the search when they are combined with other methods. As the detection of these stages is hard (or even impossible), the simple learning mechanism used in the portfolio is able to properly detect which algorithm should be applied at every stage.
- Given that the portfolio includes three algorithms from the literature that have shown a good performance in the DMCLP, we can consider the portfolio as a high-performance method for this problem.

In short, we can conclude that the algorithm portfolio is a suitable tool for dealing with the DMCLP. These results make evident that the algorithm portfolio deserves greater attention and that there is still room for improvement. For example, further analysis should be conducted on the portfolio composition and the learning scheme. Another line of research can be focused on the DMCLP considering uncertainty on some parameters like the coverage radius or the distances between the demand nodes and the facilities.

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